Chapter 3: The Cox regression model — Exercises

STA310 Survival Analysis (Master's in Applied Biostatistics)

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5.1 Fitting a Cox model in R 5

Chapter 3 introduces the Cox proportional hazards model, the central modelling tool for relating survival experience to explanatory variables. These exercises cover the formulation of the model, its fitting via partial likelihood, inference for hazard ratios, model comparison and selection, and the interpretation and use of the fitted model. Model checking (residuals and the proportional hazards assumption) is taken up in Chapter 4.

1 The Cox regression model (Book §3.1–3.2)

1.1 The hazard ratio

Type: Concept & notation

For two treatment groups, the proportional hazards model is $h_N(t) = \psi\, h_S(t)$, where $h_N(t)$ and $h_S(t)$ are the hazards for the new and standard treatments.

(a) What does the quantity ψ represent, and what does the value $\psi = 0.7$ mean in words?

(b) Why is ψ re-expressed as $\exp(\beta)$? What is β called, and what sign of β corresponds to the new treatment being inferior?

(c) For $\beta = -0.3$, compute the hazard ratio and state which treatment is better.

1.2 The baseline hazard and the linear component

Type: Concept & notation

The general Cox model for the hazard of the ith individual is

$$h_i(t) = \exp(\beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_p x_{pi})\, h_0(t).$$

(a) What is $h_0(t)$, and for whom is it the hazard?
(b) Why does the linear component contain no intercept (constant) term β_0?
(c) The quantity $\eta_i = \beta_1 x_{1i} + \cdots + \beta_p x_{pi}$ has two other names. What are they, and how is η_i used to compare two individuals?

1.3 Fitting a factor

Type: From research question to model

A study of survival after kidney cancer (hypernephroma) records two factors: age group (< 60, 60–70, > 70) and whether a nephrectomy was performed (no / yes).

(a) How many indicator (dummy) variables are needed to fit age group as a factor, and why? How many degrees of freedom does the factor have?
(b) Taking age group < 60 and "no nephrectomy" as the baseline levels, describe in words the patient to whom the baseline hazard $h_0(t)$ corresponds.
(c) Write the model for the hazard of a patient aged > 70 who had a nephrectomy, in terms of the relevant coefficients.

1.4 Proportional hazards and the semi-parametric model

Type: Assumptions

Builds on: Chapters 1–2

(a) In your own words, state the proportional hazards assumption underlying the Cox model, and say what kind of behaviour it rules out.
(b) The baseline hazard $h_0(t)$ is left completely unspecified — no distributional assumption is made about it. Why is the Cox model therefore called semi-parametric?
(c) Why does leaving $h_0(t)$ unspecified still allow the β-coefficients (and hence the hazard ratios) to be estimated?

2 Fitting the model and making inferences (Book §3.3–3.5)

2.1 The partial likelihood and the risk set

Type: Risk set reasoning

Builds on: Chapter 2

Five patients are followed. Patients 2 and 5 are right-censored, and the three deaths, in order, are patients 3, 1 and 4 at times $t_{(1)} < t_{(2)} < t_{(3)}$. Patient 5 is censored before $t_{(2)}$, and patient 2 between $t_{(2)}$ and $t_{(3)}$. Write $\psi(i) = \exp(\beta' x_i)$.

(a) List the risk set at each of the three death times.
(b) Write the partial likelihood as a product over the three death times.
(c) Censored patients contribute no term to the numerator of the partial likelihood. Where does their information enter, and what does this imply about which feature of the data drives the estimate of β?

2.2 Interpreting Cox regression output

Type: Interpret R output

For the breast cancer data (Example 1.2), a Cox model with staining indicator X (1 = positive, 0 = negative) gives $\hat\beta = 0.908$ with $\mathrm{se}(\hat\beta) = 0.501$.

(a) Compute the estimated hazard ratio and interpret it in words.
(b) Compute a 95% confidence interval for the log hazard ratio β, and hence for the hazard ratio. Why is the interval obtained by exponentiating the β-limits preferable to $\hat\psi \pm 1.96\,\mathrm{se}(\hat\psi)$?
(c) The null model has $-2\log \hat L = 173.968$ and the model with X has 170.096. Carry out the likelihood-ratio test and interpret. (The corresponding log-rank test gave $\chi^2 = 3.515$, $P = 0.061$.)

2.3 Interpreting a multi-variable model

Type: Interpret R output

For the multiple myeloma data (Example 1.3), a Cox model is fitted with all seven explanatory variables. Of the individual 95% confidence intervals, only those for Bun ($\hat\beta = 0.021$, se $= 0.006$) and Hb ($\hat\beta = -0.135$, se $= 0.069$) exclude zero. When Bun is omitted, the estimated coefficient of Hb changes from -0.135 to -0.140, and the other coefficients change too.

(a) Why can we *not* conclude from these individual results alone that only Bun and Hb are needed in the model?
(b) What does the change in the estimated coefficients when Bun is omitted tell us about the estimates?
(c) In a model with several variables, what does it mean to say that a coefficient is "adjusted for" the others?

2.4 Comparing nested models

Type: Model comparison

For the hypernephroma data (Example 3.4), the values of $-2\log\hat L$ for five nested models are summarised below. Age group is fitted via A_2, A_3 and nephrectomy via N; the interaction uses the products A_2N, A_3N.

Terms in model	Variables	$-2\log\hat{L}$
none	—	177.667
age group	A_2, A_3	172.172
nephrectomy	N	170.247
age group + nephrectomy	A_2, A_3, N	165.508
1. interaction	A_2, A_3, N, A_2N, A_3N	162.479

(a) Test for an interaction between age group and nephrectomy. What do you conclude?
(b) Does age group add to a model that already contains nephrectomy? Does nephrectomy add to a model that already contains age group?
(c) Which model would you select, and why? What role do the degrees of freedom play in these comparisons?

3 Model selection (Book §3.6–3.8)

3.1 A strategy for model selection

Type: Method selection

(a) State the hierarchic principle, and give an example of a situation where it applies.
(b) Why does the book recommend against relying uncritically on automatic forward selection, backward elimination or stepwise procedures?
(c) Describe the first step of the recommended model-selection strategy, and explain why a relatively large significance level (about 15%) is suggested for guiding inclusion.

3.2 Variable selection with the lasso

Type: Method selection

(a) What does the lasso do to the estimated coefficients, and how does it carry out variable selection?
(b) Why must the explanatory variables be standardised before the lasso is applied?
(c) How is the optimal value of the tuning parameter λ chosen, and why are standard errors and hazard ratios usually not reported after using the lasso?

4 Interpreting and using the fitted model (Book §3.9–3.11)

4.1 Adjusted and unadjusted hazard ratios

Type: What's wrong here?

For the prostatic cancer data (Example 1.4), a model containing only the treatment indicator gives a coefficient of -1.978 (hazard ratio 0.14). After adjusting for tumour size and Gleason index, the coefficient is -1.113 (hazard ratio 0.329).

(a) Explain in words what the two hazard ratios estimate, and why they differ.

(b) A researcher reports "treatment reduces the hazard of death by 86%", based on the unadjusted model. What is wrong with this statement?
(c) Which hazard ratio is the better summary of the treatment effect, and why?

4.2 Estimating the survivor function

Type: Concept & notation

Builds on: Chapter 2

After fitting a Cox model, the estimated survivor function for an individual is

$$\hat{S}_i(t) = \left\{ \hat{S}_0(t) \right\}^{\exp(\hat{\beta}' x_i)}.$$

(a) What is $\hat{S}_0(t)$, and how is $\hat{S}_i(t)$ obtained from it?
(b) With no covariates in the model, show that this formula reduces to the Kaplan–Meier estimate.
(c) In the hypernephroma example, nephrectomy has coefficient -1.412. If the baseline survivor function at 12 months is $\hat{S}_0(12) = 0.486$, estimate the 12-month survival probability for a patient aged < 60 who had a nephrectomy, and explain why the two estimated curves cannot cross.

5 Mini-lab: fitting a Cox model in R

5.1 Fitting a Cox model in R

Type: Mini-lab

Run the code below. It simulates 200 patients with a treatment indicator (`treat`) and age, generates survival times with a proportional-hazards structure, censors them, and fits a Cox model.

```
set.seed(2718)
n <- 200
treat <- rep(c(0, 1), each = n/2)
age <- round(runif(n, 45, 80))
eta <- 0.55 * treat + 0.04 * (age - 60)
t <- rexp(n, rate = 0.08 * exp(eta))
cens <- runif(n, 0, 60)
obs <- pmin(t, cens)
status <- as.numeric(t <= cens)
d <- data.frame(time = obs, status = status, treat = treat, age = age)

library(survival)
fit <- coxph(Surv(time, status) ~ treat + age, data = d)
summary(fit)
```

```
Call:
coxph(formula = Surv(time, status) ~ treat + age, data = d)
```

```
  n= 200, number of events= 172

          coef exp(coef) se(coef)     z Pr(>|z|)
treat 0.641981  1.900242 0.162533 3.950 7.82e-05 ***
age   0.037207  1.037908 0.007944 4.683 2.82e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

      exp(coef) exp(-coef) lower .95 upper .95
treat     1.900     0.5262     1.382     2.613
age       1.038     0.9635     1.022     1.054

Concordance= 0.603  (se = 0.026 )
Likelihood ratio test= 29.62  on 2 df,   p=4e-07
Wald test            = 29.56  on 2 df,   p=4e-07
Score (logrank) test = 29.97  on 2 df,   p=3e-07
```

(a) From the summary, report the estimated hazard ratio and its 95% confidence interval for treatment, and interpret.
(b) Report the hazard ratio per additional year of age, and interpret.
(c) How many events are there? What does the "Concordance" line measure, and is the model useful for discrimination?
(d) What are the three tests at the bottom of the output, and do they agree?